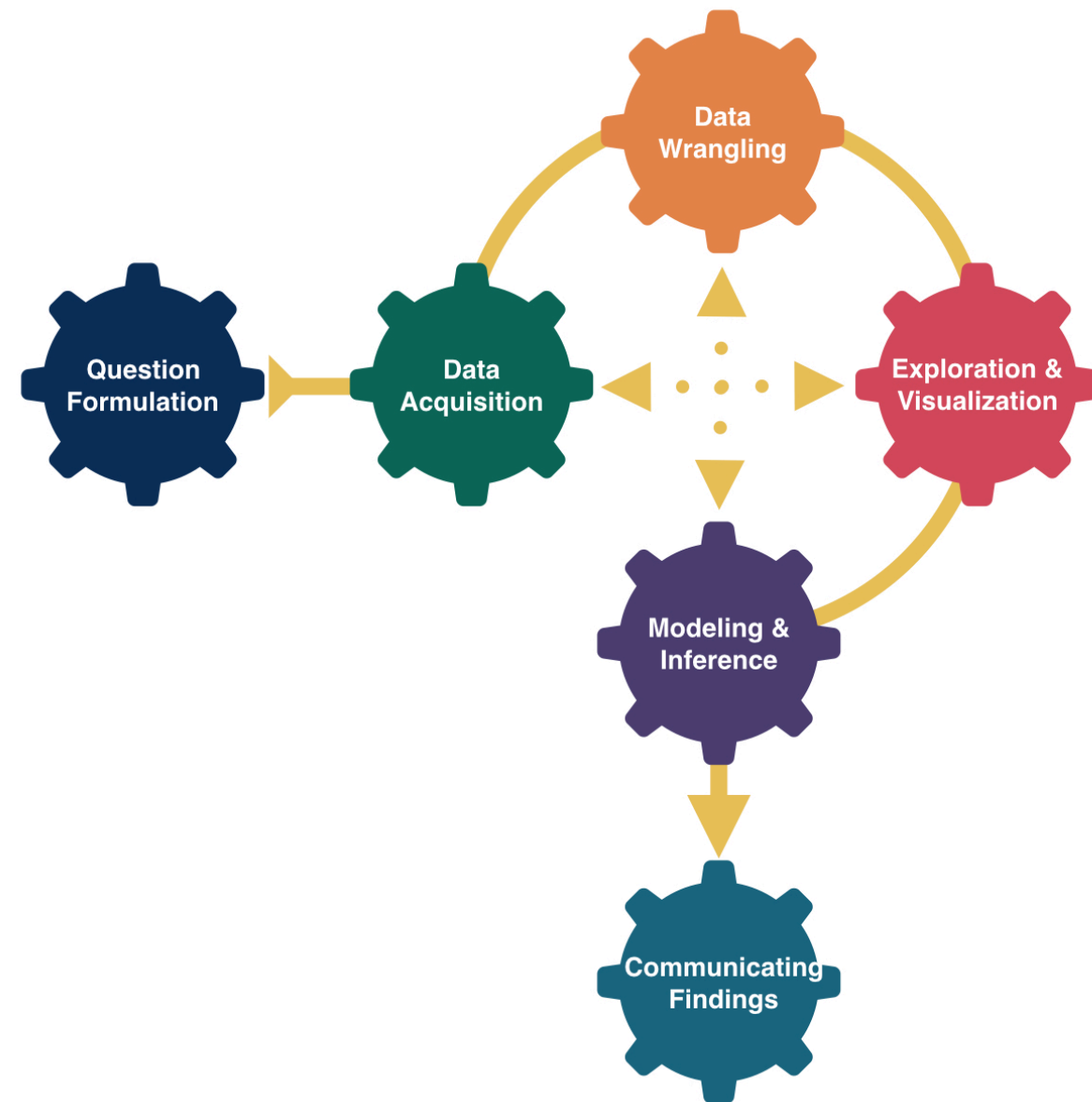


Developing Coding Taste



Kelly McConville

DATA 100

Week 14 | Spring 2026

Announcements

- Oral exam sign-ups in Slack
- **Extra Credit Lecture Quiz:** Due In-Person at 3pm (start of lab) on Friday, May 1st
- **Exam Rewrites:** Due on Gradescope at noon on Monday, March 4th

Goals for Today

- Developing Coding Taste
- Code smells
- Learning a new coding task: maps!

Recap: Vibe coding

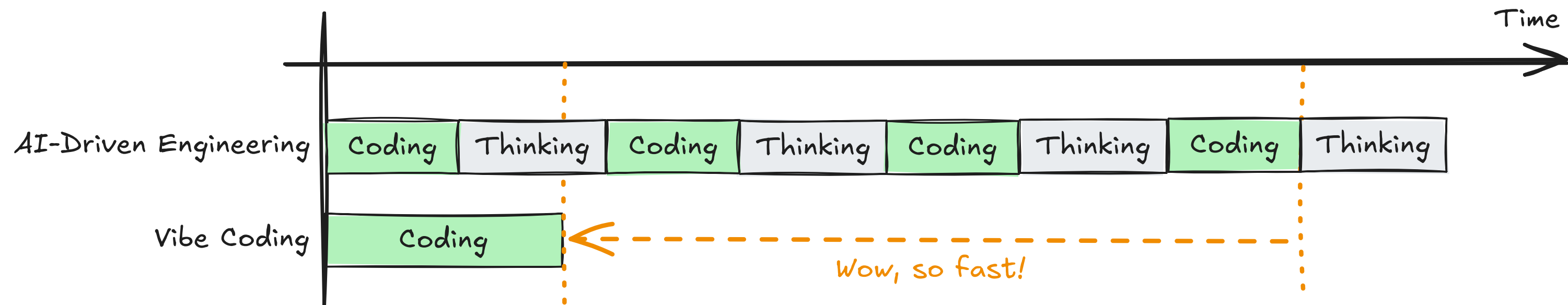
- Vibe coding is where an AI tool generates code based on prompts from a person.
- Important skills to develop:
 - Ability to review and debug the AI generated code.
 - Ability to prompt the AI tool for better code.
 - A strong understanding of what you want the AI code to produce.
 - Don't let it take you down the wrong rabbit hole.
 - **Taste!**

AI-Driven Engineering

Think first and often when coding with AI!

“AI-driven engineering: employing best practices, foregrounding human understanding of the code, moving slowly to make development sustainable.”

“Vibe coding: throwing caution to the wind and implementing at speed, sacrificing understanding for delivery velocity, hitting an eventual wall of unsalvageable, messy code.”



Developing Your Taste for Well Constructed Code

- Want code that:
 - Works.
 - Is robust.
 - Is easy to understand.
 - Is easy to contribute to.
- **Code smells**: Structures in code that violate at least one of the items above.

Example: Bike Counter

- Let's return to the Cambridge bike counter data.
- Import the **data**.

```
1 july_2019 <- read_csv("data/july_2019.csv")
2
3 # Inspect the data
4 glimpse(july_2019)
```

Rows: 192

Columns: 8

```
$ DateTime <chr> "07/04/2019 12:00:00 AM", "07/04/2019 12:15:00 AM", "07/04/2...
$ Day      <chr> "Thursday", "Thursday", "Thursday", "Thursday", "Thursday", ...
$ Date     <date> 2019-07-04, 2019-07-04, 2019-07-04, 2019-07-04, 2019-07-04,...
$ Time     <time> 00:00:00, 00:15:00, 00:30:00, 00:45:00, 01:00:00, 01:15:00,...
$ Total    <dbl> 2, 3, 2, 0, 3, 2, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, ...
$ Westbound <dbl> 2, 3, 1, 0, 2, 2, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1, ...
$ Eastbound <dbl> 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, ...
$ Occasion <chr> "Fourth of July", "Fourth of July", "Fourth of July", "Fourt...
```

- Suppose we didn't have the **Total** column and wanted to create it by adding **Westbound** and **Eastbound**.

Smelly Code

Smelly Code

```
1 july_2019$new_T=rep(5, 192)
2 for(i in 1:129){
3   july_2019$new_T[i]=july_2019[i,6]+july_2019[i,7]
4 }
```

Not Smelly Code

```
1 july_2019 <- mutate(july_2019,
2                       new_Total = Westbound + Eastbound)
```

Both are correct-ish:

```
1 glimpse(july_2019)
```

Rows: 192

Columns: 10

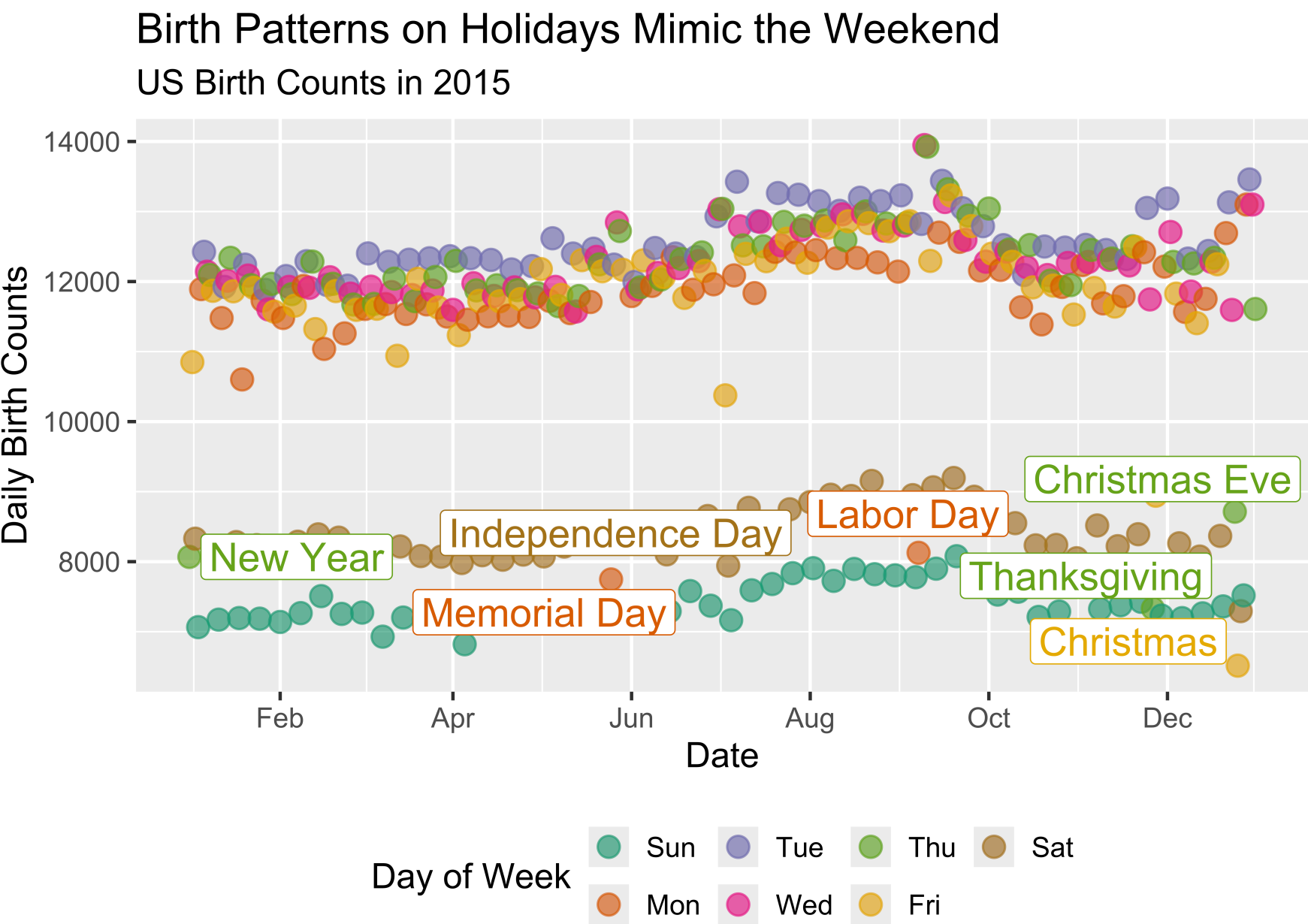
```
$ DateTime <chr> "07/04/2019 12:00:00 AM", "07/04/2019 12:15:00 AM", "07/04/2...
$ Day      <chr> "Thursday", "Thursday", "Thursday", "Thursday", "Thursday", ...
$ Date     <date> 2019-07-04, 2019-07-04, 2019-07-04, 2019-07-04, 2019-07-04,...
$ Time     <time> 00:00:00, 00:15:00, 00:30:00, 00:45:00, 01:00:00, 01:15:00,...
$ Total    <dbl> 2, 3, 2, 0, 3, 2, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, ...
$ Westbound <dbl> 2, 3, 1, 0, 2, 2, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1, ...
$ Eastbound <dbl> 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, ...
$ Occasion <chr> "Fourth of July", "Fourth of July", "Fourth of July", "Fourt...
$ new_T    <list> 2, 3, 2, 0, 3, 2, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1,...
$ new_Total <dbl> 2, 3, 2, 0, 3, 2, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, ...
```

But why is the lefthand code smelly?

Developing Your Taste for Well Constructed Outputs

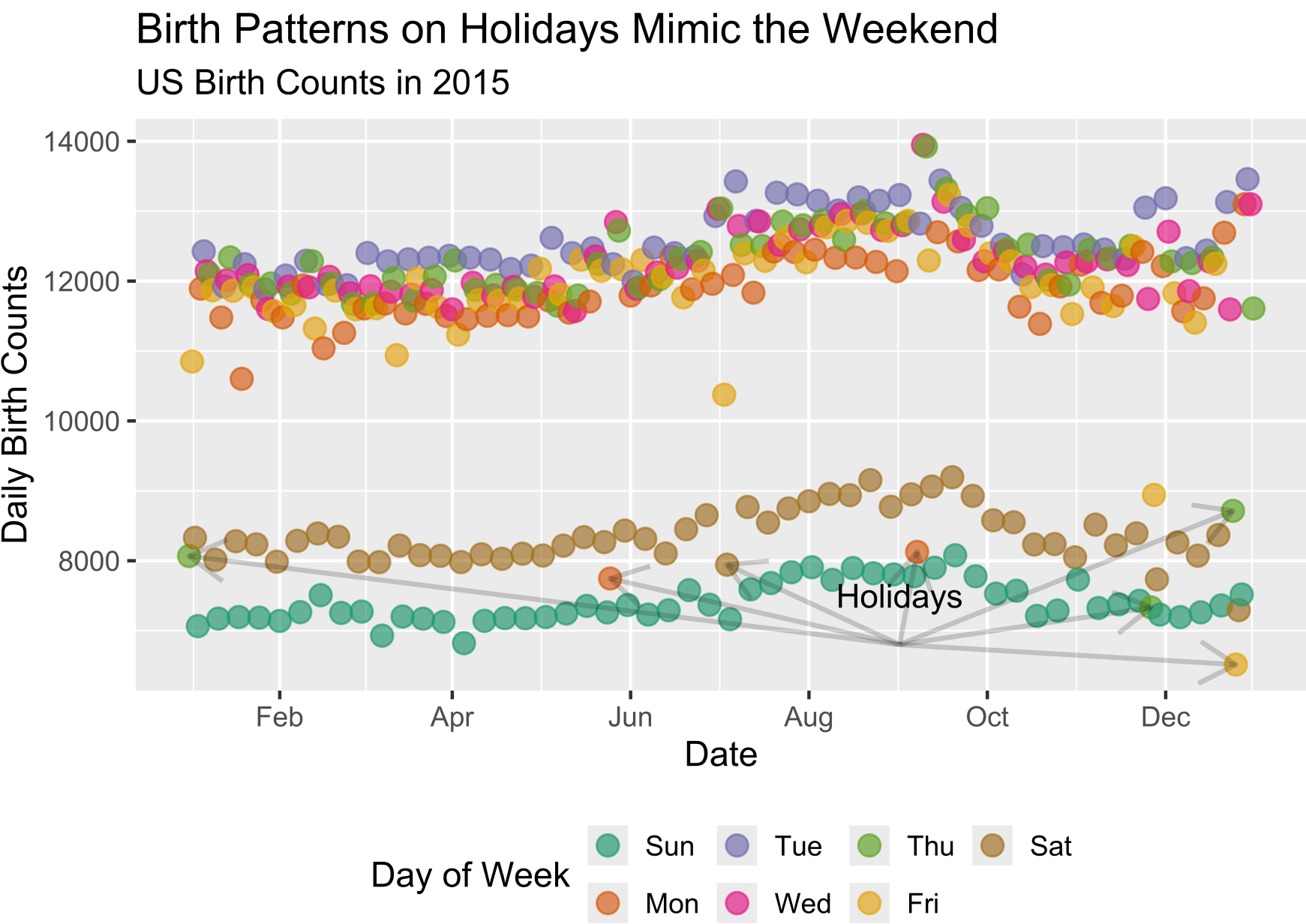
- Want outputs that are:
 - Accurate.
 - Clear.
 - Unique.
 - Provide context.
- Let's look at some of the final graphs from last class and ask:
 - Which graph do you prefer?
 - How could the graphs be improved?

Example: Birth Patterns in the US



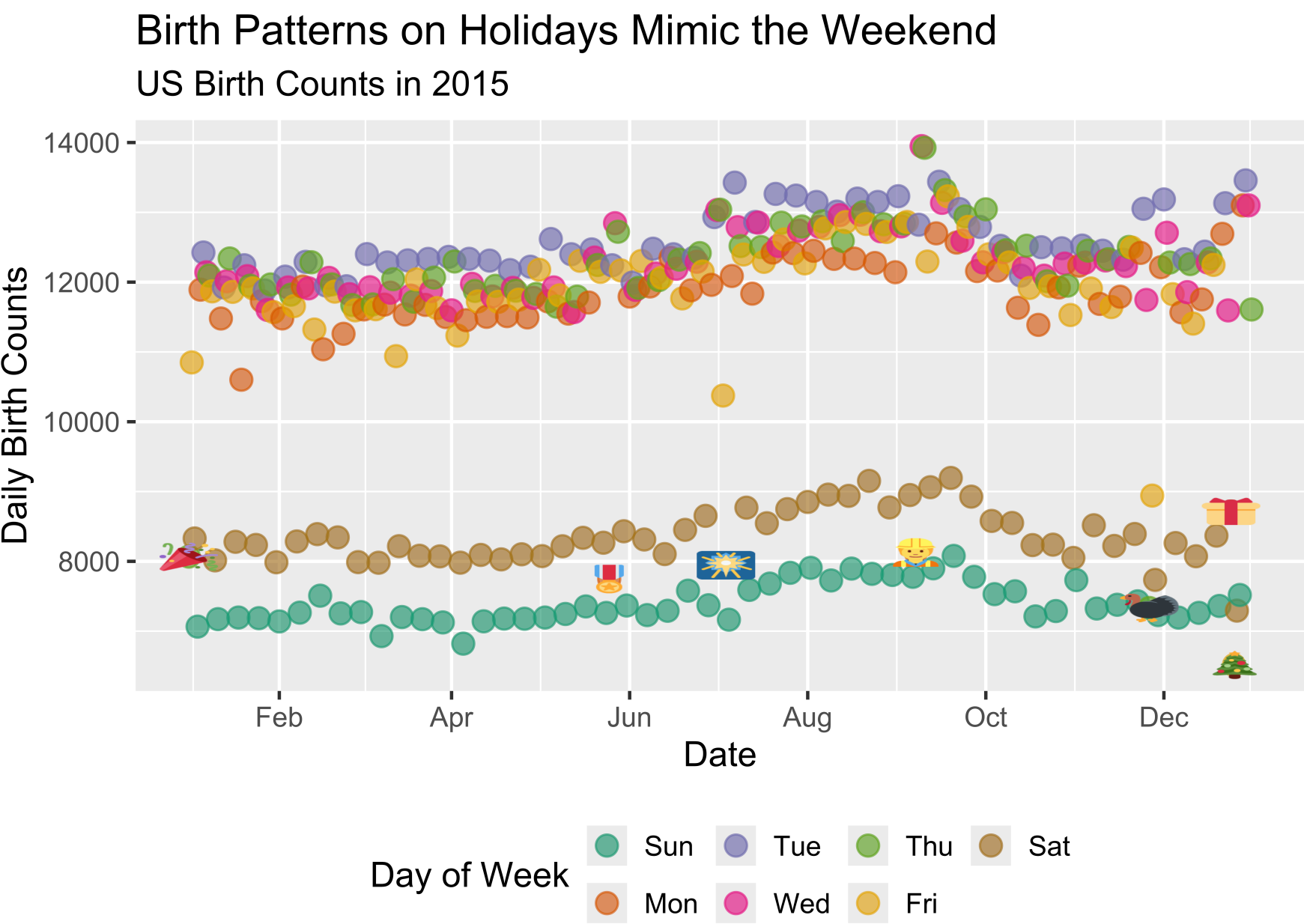
Data: National Vital Statistics System

Example: Birth Patterns in the US



Data: National Vital Statistics System

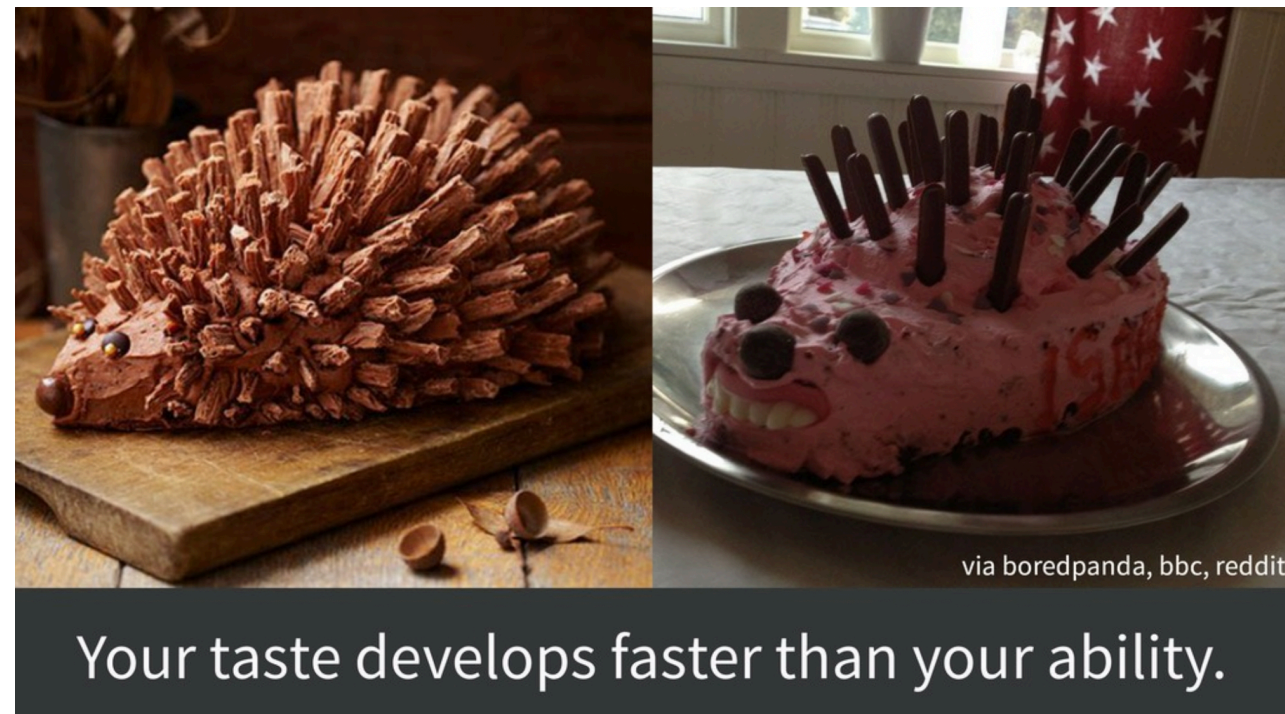
Example: Birth Patterns in the US



Data: National Vital Statistics System

Learn from Experts & AI

- [Jenny Bryan](#) is a rock star coder. Reading her materials and watching her presentations has made me a better coder.
- Jenny's observation:



- But AI can help speed up your ability by helping you learn new code.
 - As long as you keep **developing your own taste**.

Suggestions for Developing Taste while Vibe Coding

- Know what you want.

I have a dataset called Births2015 with variables date and births. Give me code for a ggplot where the geom is point and date is on the x axis and births is on the y axis.

- And then **iterate**.

Suggestions for Developing Taste while Vibe Coding

- Be curious.
 - Saw lots of great curiosity on Monday!

How do I change the font in the ggplot?

◆ To change fonts in `ggplot2`, you use the `theme()` function. Specifically, you'll want to adjust the `text` element (which affects all text) or specific elements like `plot.title` or `axis.text`.

How do I find the max births for each wday?

◆ To find the maximum births for each day of the week, you'll want to use the `dplyr` package. It allows you to "group" your data by a specific variable and then perform a calculation on each of those groups.

Here is the code to calculate the max births for each `wday` :

Suggestions for Developing Taste while Vibe Coding

- Keep code concise and consistent.
 - Ex: Don't load packages at the start of every chunk!

```
1 library(dplyr)
2 library(ggplot2)
```

Suggestions for Developing Taste while Vibe Coding

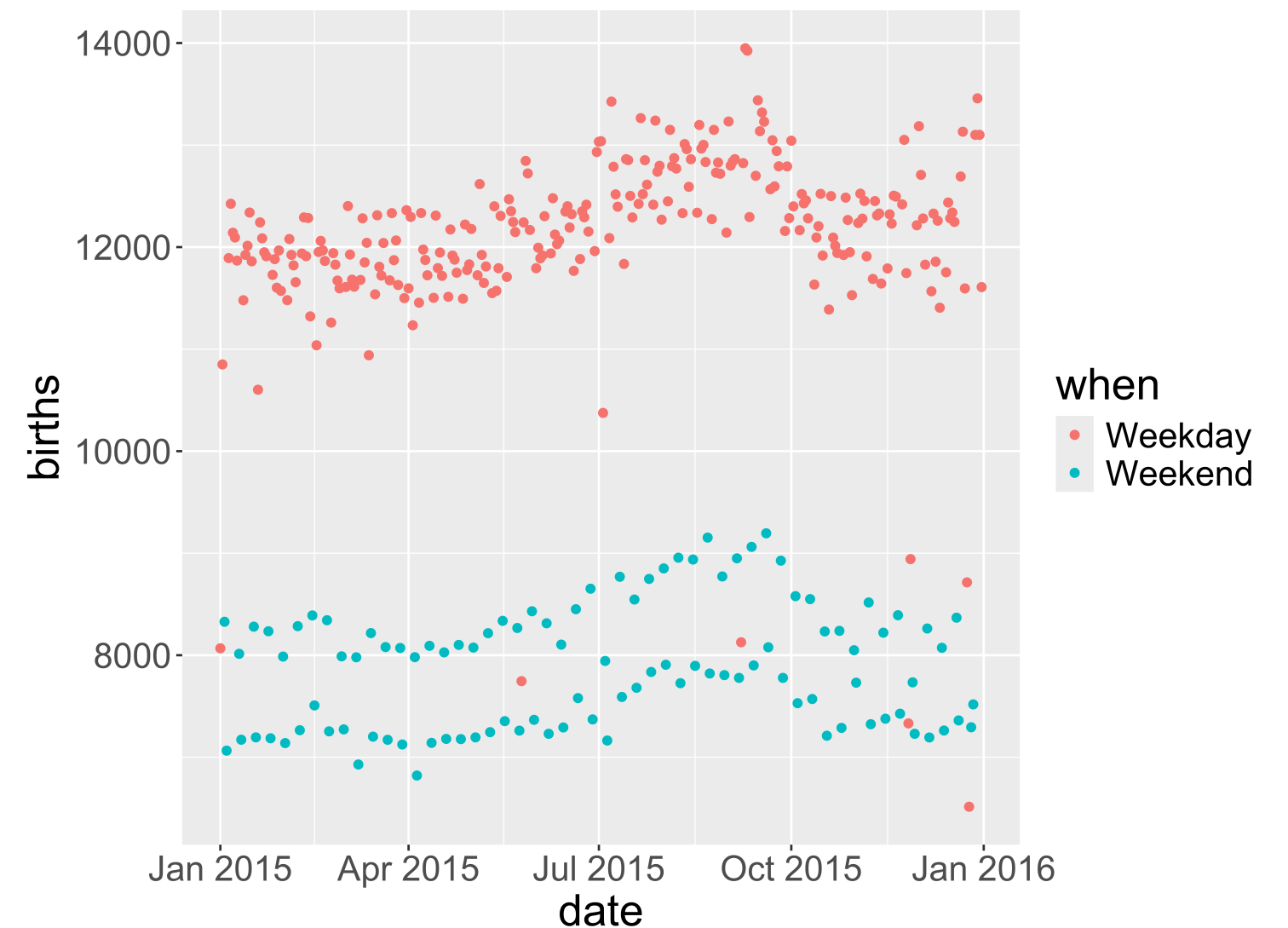
Don't over-comment your code.

```
1 # Create the plot with a qualitative ColorBrewer palette
2 ggplot(data = Births2015, aes(x = date, y = births, color = wday)) +
3   geom_point(alpha = 0.8, size = 1.5) + # Added alpha and size for better visibility
4   scale_color_brewer(palette = "Dark2") + # Applies the distinct qualitative palette
5   theme(
6     legend.position = "bottom", # Moves legend to bottom for a wider plot area
7     plot.title = element_text(face = "bold") # Bolds the title for a cleaner look
8   )
```

Suggestions for Developing Taste while Vibe Coding

Be unique and don't settle for the defaults.

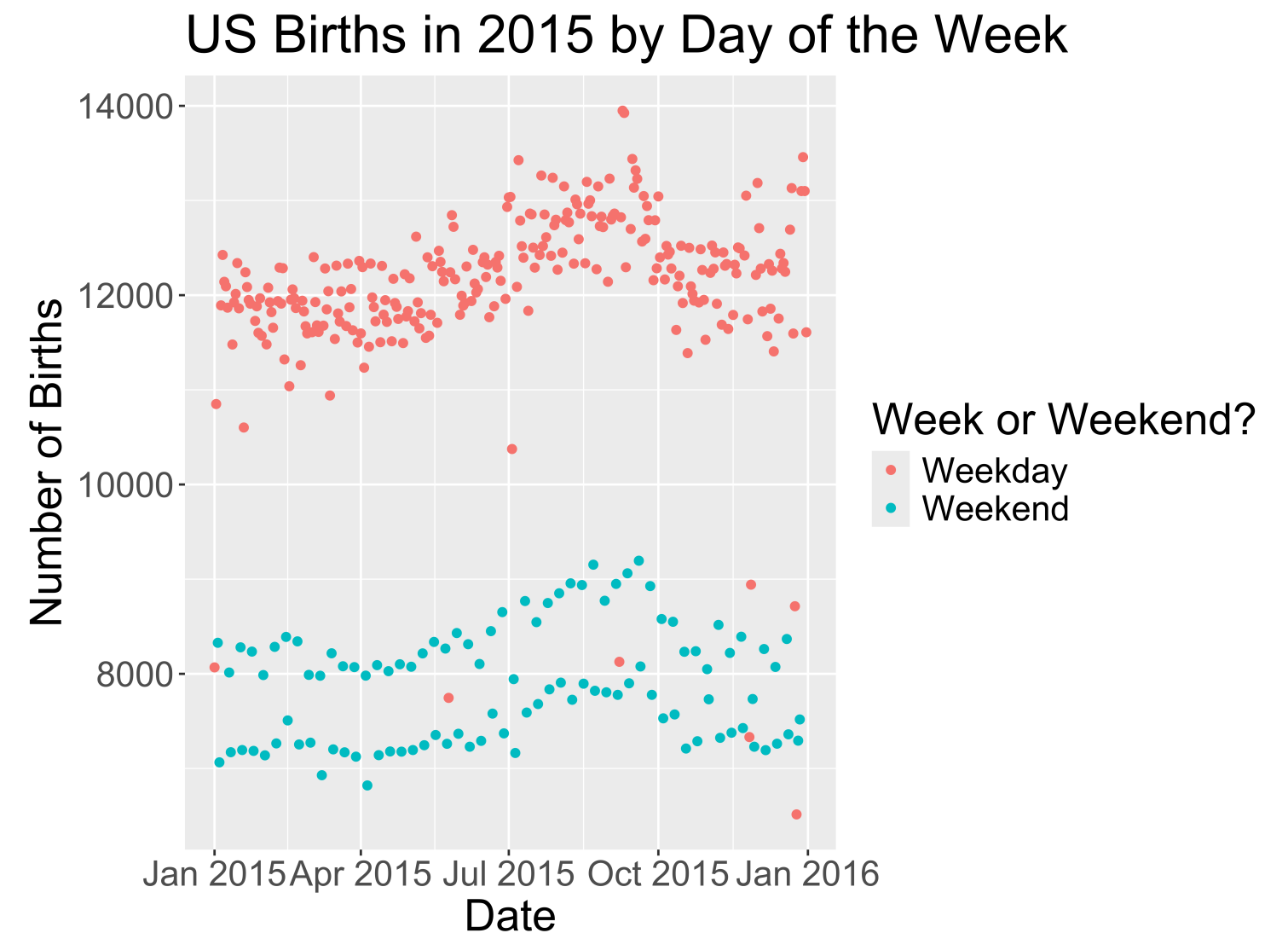
```
1 Births2015 <- mutate(Births2015, when = if_else(  
2   day_of_week %in% c(1, 7), "Weekend", "Weekday"  
3 ))  
4  
5 ggplot(data = Births2015,  
6   mapping = aes(x = date, y = births,  
7                 color = when)) +  
8   geom_point()
```



Suggestions for Developing Taste while Vibe Coding

Avoid the illusion of context.

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = when)) +  
4 geom_point() +  
5 labs(  
6   title = "US Births in 2015 by Day of the Week",  
7   x = "Date",  
8   y = "Number of Births",  
9   color = "Week or Weekend?"  
10 )
```

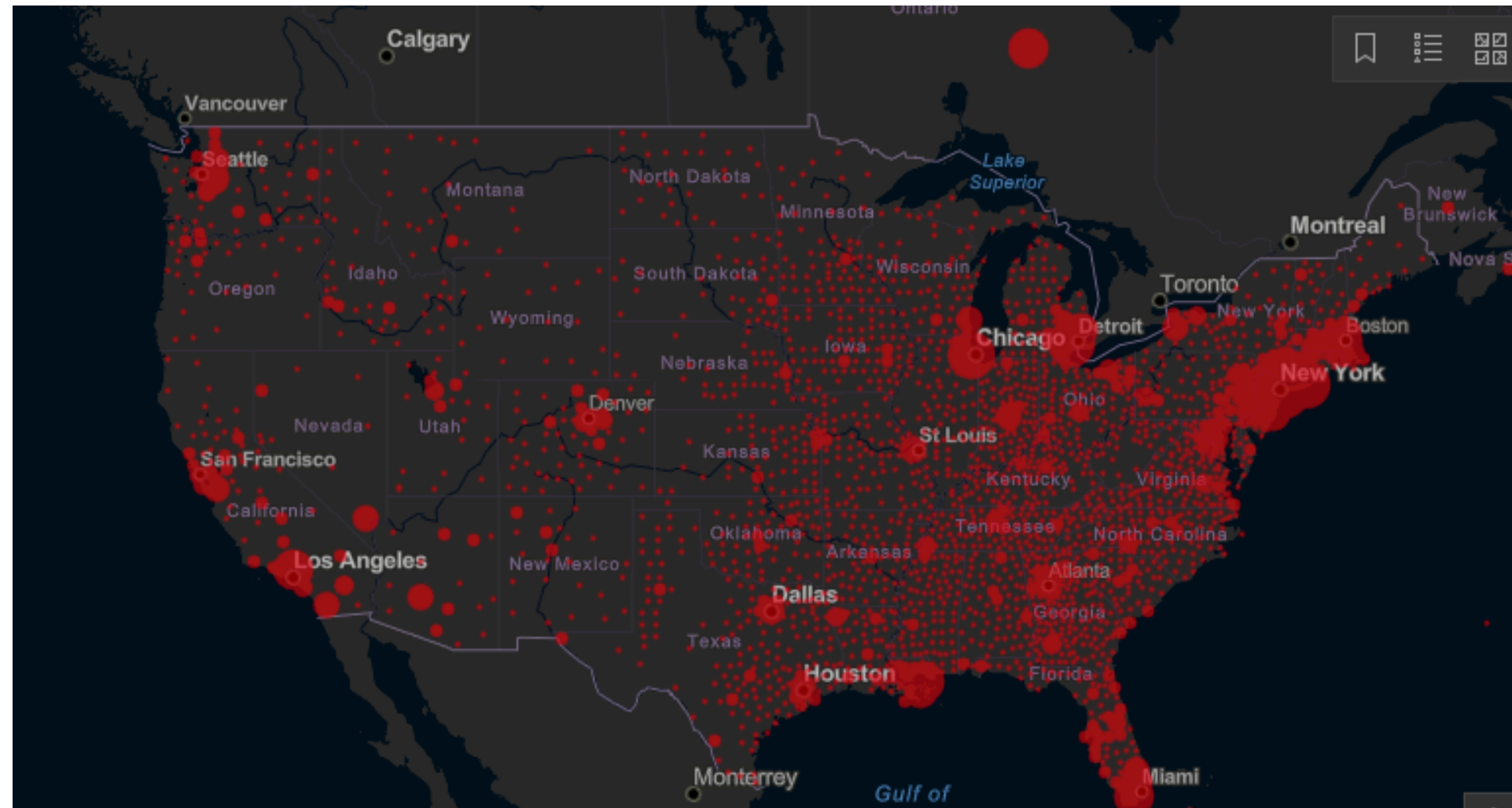


Let's learn something new: Maps in R!

Spatial Data

- Data that is linked to the physical world.
- Visualizing these data on a map (or map-like graph) is a great way to add useful context!
- Focus on choropleth maps and raster maps.

Static, Raster Maps

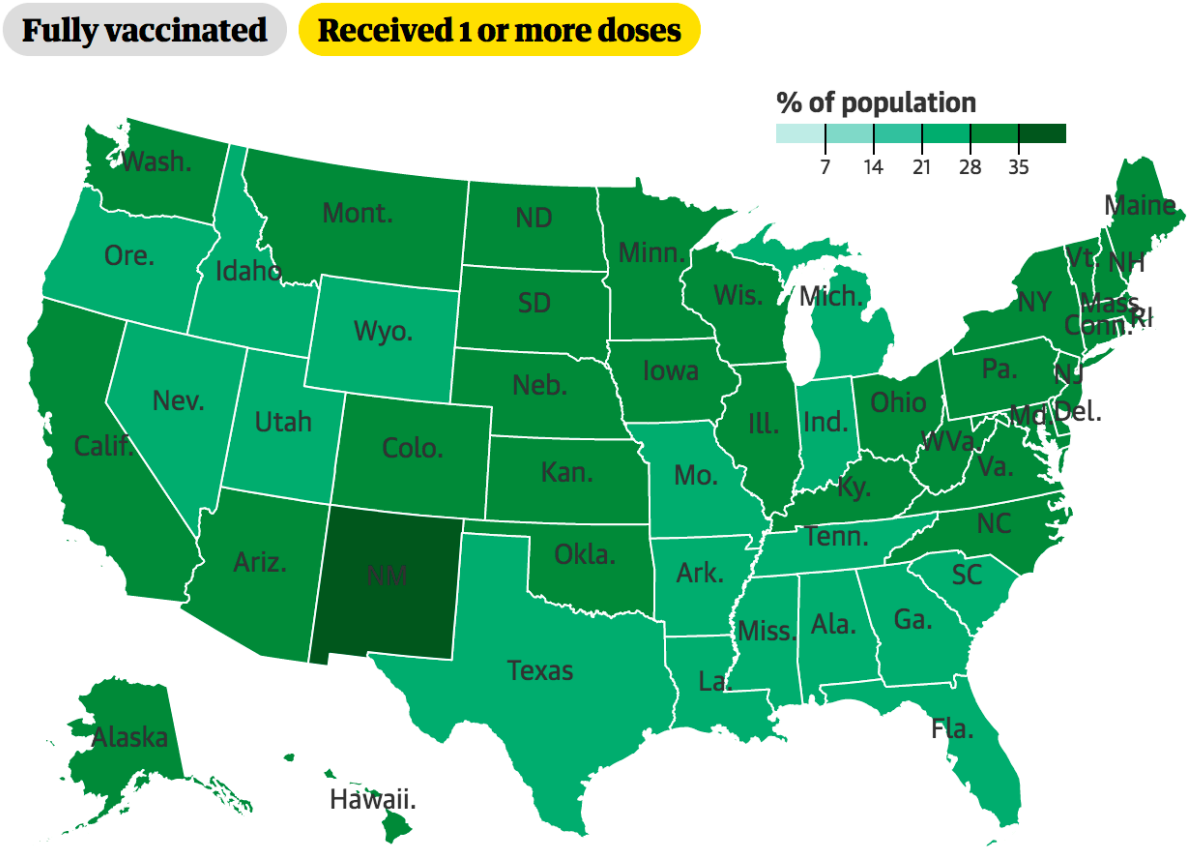


Source: JHU

Choropleth Maps

Vaccination rates across the US

Fully vaccinated (nationwide)	Received 1 or more doses (nationwide)
15.8%	28.6%



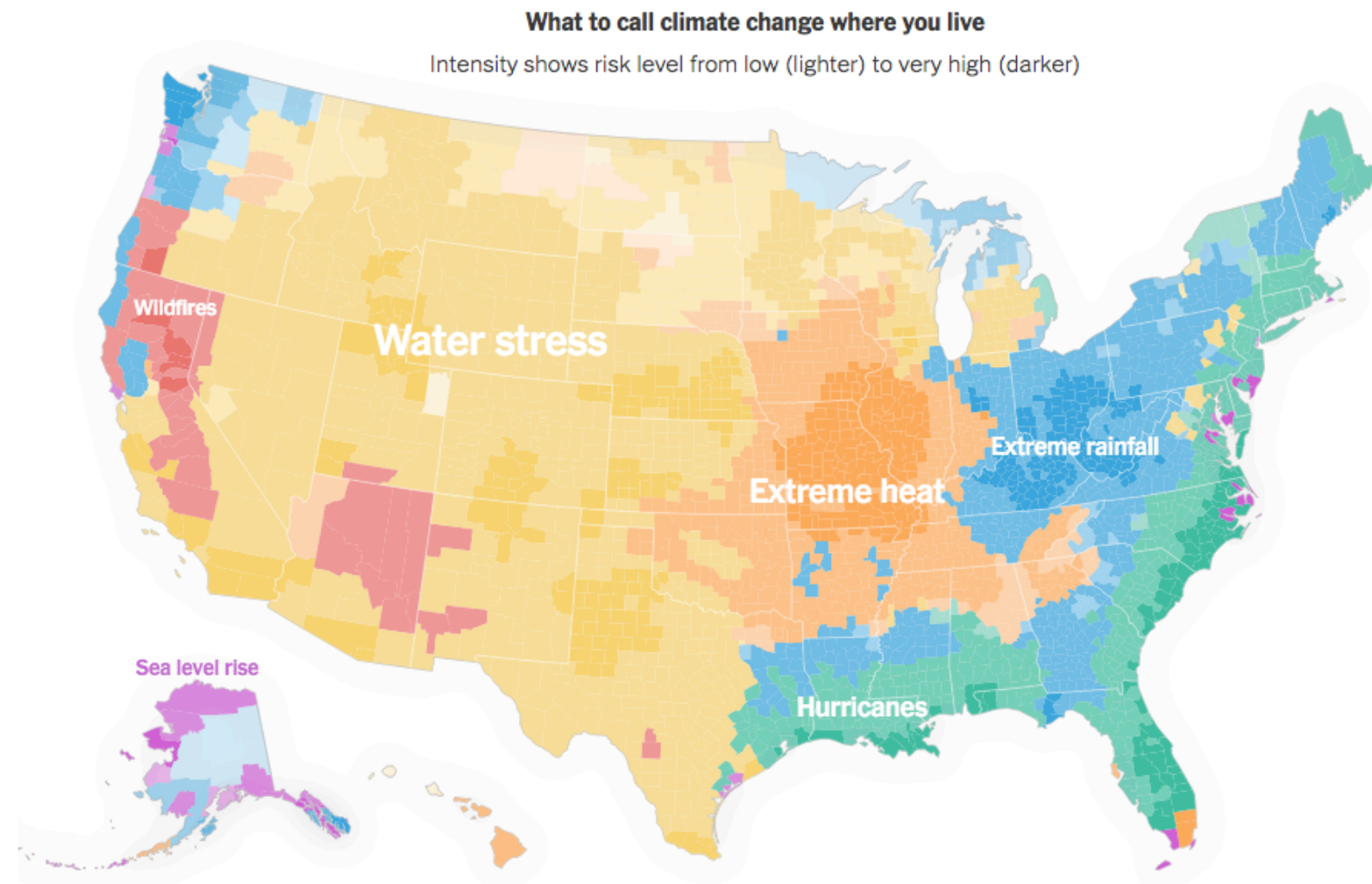
Guardian graphic | Source: Centers for Disease Control. Last updated on 2021-03-29.

Note: The percentages reported here represent a proportion of the total state population, including children. However, the Pfizer vaccine is only available to those 16 years and older, whereas the Moderna vaccine is available to those who are 18 years or older. Additionally, the CDC has updated its data since this article was first published. Single-dose vaccines are now included in those who are considered "Fully vaccinated" as well as those who have "Received 1 or more doses."

Source: The Guardian

Interactive Maps

Can be choropleth or raster maps!



Source: [NYTimes](#)

Spatial Data Structures

- Often stored in special data structures
 - Ex: Shapefile(s)
 - Consist of geometric objects like points, lines, and polygons
- Visualizing Spatial Data
 - Need to draw the map
 - Add metadata on top of the map
 - Color in regions
 - Dots

Downloading Materials from Posit Cloud

Let's Learn Something New: Mapping Activity in Posit Cloud

